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Project Proposal

NSTT-Lite: Auditing and Repairing a Published Nepali ASR Model

Module: ST7088CEM — Artificial Neural Networks

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Problem

`gagan3012/wav2vec2-xlsr-nepali` is a published Hugging Face XLS-R (wav2vec2) model for Nepali speech recognition that self-reports **5.97% WER** — a remarkably strong number for a low-resource language. However, that figure was measured on the model’s own training corpus, OpenSLR-43, which is effectively single-speaker (one female voice). My preliminary experiments confirm the claim technically holds in-domain (I measured **4.91% WER** on OpenSLR-43) but the same model collapses to roughly **65% WER** on multi-speaker Nepali speech (OpenSLR-54) — a >13x degradation. The published benchmark therefore does not describe real-world performance. This project audits that claim rigorously and then repairs the model.

Tasks

1. **Benchmark audit.** Reproduce the self-reported figure in-domain (OpenSLR-43) and measure the same checkpoint zero-shot on a speaker-disjoint multi-speaker test split (OpenSLR-54), quantifying the generalization gap.
2. **Fine-tuning.** Fine-tune the same XLS-R model (CTC objective) on a ~15-hour, 160-speaker, speaker-disjoint OpenSLR-54 training subset to close that gap.
3. **Generalization re-evaluation.** Evaluate original vs. fine-tuned checkpoints on both test sets — including a catastrophic-forgetting check on the original corpus — plus error analysis and a speaker-leakage ablation.

Datasets

- **OpenSLR-54** — crowdsourced multi-speaker Nepali ASR corpus (Kjartansson et al., SLTU 2018), 157,905 utterances, CC BY-SA 4.0. Link: <https://www.openslr.org/54/> (Working subset already prepared: 15,171 utterances / 160 speakers / 15.03 hours, speaker-disjoint 80/10/10 split, zero speaker overlap.)
- **OpenSLR-43** — the model’s own training corpus (single-speaker female Nepali TTS-style data), used only for in-domain reproduction of the claim. Link: <https://www.openslr.org/43/>

The two corpora are kept strictly separate in all results.

Method and infrastructure

Fine-tuning uses the Hugging Face `transformers` CTC training stack on a Colab T4 GPU (FP16, batch 2, gradient accumulation, early stopping on validation WER). All experiments — audit runs, training, and re-evaluation — are tracked with **MLflow** (parameters, WER/CER metrics, artifacts), giving a reproducible evidence trail; environment versions are pinned and device screenshots captured throughout.

Work plan (10 phases)

Phase	Deliverable
1	This proposal
2	Environment + MLflow reproducibility setup

Phase	Deliverable
3	Data preparation (complete: speaker-disjoint 15hr subset)
4	Baseline audit: in-domain vs. out-of-domain zero-shot WER
5	Algorithm selection justification
6	XLS-R fine-tuning on speaker-diverse data (Colab T4)
7	Generalization re-evaluation (4-cell before/after $\times$ in/out-of-domain)
8	Efficiency benchmark (latency, size, int8 quantization)
9	Error analysis, leakage ablation, discussion
10	Report writing, evidence assembly, submission

Achievability

The audit half is already demonstrated end-to-end at small scale (the 4.91% / ~65% preliminary numbers above), and the data pipeline (download, preprocessing, speaker-disjoint splitting) is built and validated. The main remaining cost is the Phase 6 GPU fine-tuning run, which fits a free Colab T4 budget with checkpointed, resumable training. Every phase produces a concrete, checkable artifact (a metric, a checkpoint, an MLflow run), so progress is verifiable throughout rather than only at submission.